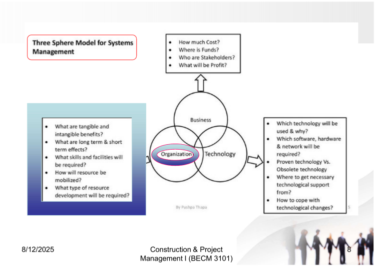
CONSTRUCTION AND PROJECT MANAGEMENT - I

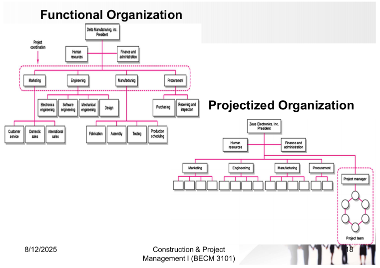
Section B - Lecture 02

PROJECT ORGANIZATIONAL CONCEPT & STRUCTURE

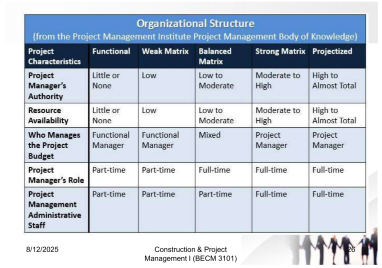
Term-Final Question-to-Slide Mapping | 2016-2024



Three Sphere Model



Functional vs projectized



Organizational Structure
(from the Project Management Institute Project Management Body of Knowledge)

Project Characteristics	Functional	Weak Matrix	Balanced Matrix	Strong Matrix	Projectized
Project Manager's Authority	Little or None	Low	Low to Moderate	Moderate to High	High to Almost Total
Resource Availability	Little or None	Low	Low to Moderate	Moderate to High	High to Almost Total
Who Manages the Project Budget	Functional Manager	Functional Manager	Mixed	Project Manager	Project Manager
Project Manager's Role	Part-time	Part-time	Full-time	Full-time	Full-time
Project Management Administrative Staff	Part-time	Part-time	Part-time	Full-time	Full-time

Authority/resource comparison

Scope: The 33-slide lecture was checked against all term-final papers from 2016 through 2024. Original section labels are preserved, while topic classification follows the lecture because Section A/B placement changed across years.

Match levels: Exact = almost the complete question is answerable from the lecture; Direct = the main demand is covered but requires judgment/synthesis; Partial = only one clause or a supporting part is present.

1. Executive Summary

17	9	6	2	186
Question-to-slide links	Exact matches	Direct matches	Partial matches	Total linked marks
All nine exam years	98 linked marks	64 linked marks	24 linked marks	Full marks of linked parts

Most repeated area: Functional, projectized and matrix organization - definitions, differences, authority, advantages/disadvantages and selection. These topics appear in every exam year.

Year	Linked marks	Relative coverage
2016	26	■■■■■■■■■■■■■■■■■■■■■■■■■■
2017	7	■■■■■■■
2018	13	■■■■■■■■■■■■■
2019	12	■■■■■■■■■■■■
2020	10	■■■■■■■■■■
2021	35	■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
2022	37	■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
2023	23	■■■■■■■■■■■■■■■■■■■■■■■
2024	23	■■■■■■■■■■■■■■■■■■■■■■■

Recurring theme	Main slides	Observed pattern
Project organization, challenges and critical issues	L2-6	Asked in 2016, 2017, 2018, 2020, 2022 and 2024.
Three basic structures and pros/cons	L9-26, L32-33	Appears in 2016, 2019, 2021, 2022, 2023 and 2024.
Choosing an appropriate structure	L30-31	Explicitly tested in 2017, 2018, 2020, 2021 and 2024.
System view / Three Sphere Model	L7-8	A focused 12-mark question in 2021.
Top-down and bottom-up design	L27-29	Top-down design appeared as part of a 2016 question.

2.1 Detailed Question-to-Slide Mapping: 2016-2020

Year	Original Sec./Q	Marks	Condensed question	Lecture slide(s)	Match	Coverage assessment
2016	A 1(c)	8	Define project organization and discuss the different types of project organization.	L3-6, L9-26	Exact	The lecture defines organization and develops functional, projectized and matrix structures.
2016	A 1(d)	9	Write the pros and cons of functional, projectized and matrix organization.	L12-13, L19, L24, L32-33	Exact	Advantages and disadvantages of all three structures are explicitly listed.
2016	A 4(b)	9	Conceptual design process and top-down design style with example.	L27-28	Partial	Top-down design and its example are present; the conceptual-design process belongs to another lecture.
2017	A 3(c)	7	Define project organization, state its critical issues and select the best organization for a building project.	L3-6, L9-31	Direct	Definition, issues, structures and selection factors are supplied; the final choice requires project-specific judgment.
2018	B 6(a)	13	Define project organization, explain critical issues and identify the structure best suited to a building project.	L3-6, L9-31	Direct	The lecture provides the complete decision framework but does not prescribe one universal structure.
2019	A 2(b)	12	Why should a project be organized? Analyze the pros and cons of matrix and projectized organization.	L2-6, L15-24	Direct	Organizing challenges and both structures are covered; the why portion must be synthesized.
2020	A 3(b)	10	Define project organization, state critical issues and choose the best structure for a building project.	L3-6, L9-31	Direct	A repeated selection question supported by the structure and consideration slides.

QB = original question-bank PDF page. Slide numbers refer to the uploaded 33-page lecture PDF.

2.2 Detailed Question-to-Slide Mapping: 2021-2024

Year	Original Sec./Q	Marks	Condensed question	Lecture slide(s)	Match	Coverage assessment
2021	A 2(a)	12	Discuss the system view of project management and illustrate the Three Sphere Model.	L7-8	Exact	Systems philosophy, analysis, management and the business-organization-technology model are shown.
2021	A 2(b)	13	Discuss the three basic organizational structures with pros and cons.	L9-26, L32-33	Exact	The lecture gives descriptions, charts, comparisons and advantages/disadvantages.
2021	A 2(c)	10	How should an appropriate project-management structure be selected?	L2, L30-31	Exact	Project and organizational considerations are directly listed.
2022	A 2(a)	10	Define project organization; state major organizing challenges and critical issues.	L2-5	Exact	Challenges, hierarchy, policy and the three critical questions appear together.
2022	A 2(b)	12	Differentiate functional vs projectized and projectized vs matrix organizations.	L10-26	Exact	Authority, resources, reporting, staffing and comparative diagrams support both contrasts.
2022	A 3(b)	15	Define project team and teamwork; explain purposes, functions and multiple-team work in an overlapping environment.	L16-24	Partial	Dedicated teams and matrix overlap are illustrated, but formal teamwork theory and purposes are not developed.
2023	A 2(a)	12	A project manager is to receive final authority: choose the organization and state its advantages and disadvantages.	L15-19	Exact	This is the projectized structure, where the project manager has final authority; pros and cons are listed.
2023	A 2(c)	11	Differentiate matrix types and compare functional and matrix organization.	L20-26	Direct	Weak, balanced and strong matrices are shown; functional-matrix comparison is supported by the authority/resource table.
2024	A 2(a)	11	Define project organization; give selection considerations, purposes and challenges of organizational structure.	L2-5, L30-31	Direct	Definition, challenges and selection criteria are explicit; purposes are inferred from hierarchy, authority and coordination.
2024	A 2(b)	12	Write the pros and cons of functional, projectized and matrix organization.	L12-13, L19, L24, L32-33	Exact	A direct repeat of the three-structure comparison.

QB = original question-bank PDF page. Slide numbers refer to the uploaded 33-page lecture PDF.

3. Slide-to-Question Reverse Index

Slides	Core topic	Mapped questions	Priority
L2-6	**Organizing challenges, hierarchy, policy, critical issues and separation/integration approaches**	2016 A1(c); 2017 A3(c); 2018 B6(a); 2019 A2(b); 2020 A3(b); 2022 A2(a); 2024 A2(a)	Very high
L7-8	**System view and Three Sphere Model**	2021 A2(a)	Focused
L9-14	**Functional organization, charts, authority, advantages and disadvantages**	2016 A1(c-d); 2021 A2(b); 2022 A2(b); 2024 A2(b)	Very high
L15-19	**Projectized organization and dedicated teams**	2016 A1(c-d); 2019 A2(b); 2021 A2(b); 2022 A2(b), A3(b); 2023 A2(a); 2024 A2(b)	Highest
L20-26	**Matrix structure, matrix types, responsibilities and comparison table**	2016 A1(c-d); 2019 A2(b); 2021 A2(b); 2022 A2(b), A3(b); 2023 A2(c); 2024 A2(b)	Highest
L27-29	**Top-down and bottom-up design styles**	2016 A4(b)	Secondary
L30-31	**How to choose an appropriate structure**	2017 A3(c); 2018 B6(a); 2020 A3(b); 2021 A2(c); 2024 A2(a)	Very high
L32-33	**Summary comparison of all structures**	2016 A1(d); 2021 A2(b); 2024 A2(b)	High

Fastest revision route: L9-26 -> L2-6 -> L30-31 -> L32-33 -> L7-8 -> L27-29.

4. Exam-Focused Answer Construction Guide

1. Project organization and critical issues

Define organization as management hierarchy/reporting and authority (L3). Explain owner policy and lifecycle division (L4). State the three critical questions (L5), then separation vs integration (L6).

2. Functional organization

Explain departmental execution, functional-manager authority and suitability for smaller projects (L10-11). Add the organization chart and the complete pros/cons (L12-14).

3. Projectized organization

State final PM authority, direct reporting and self-contained resources (L15). Explain dedicated teams (L16-18), then list pros/cons from L19.

4. Matrix organization

Define the hybrid structure and two-boss relationship (L20). Explain weak/balanced/strong matrices using L21, responsibilities using L23, and pros/cons using L24.

5. Choosing the structure

Start with the balance principle (L2). Use project factors - size, strategic importance, novelty, integration, external complexity, budget/time and resource stability (L30) - plus organizational factors (L31).

6. System view / Three Sphere

Explain systems philosophy, analysis and management (L7). Draw the business-organization-technology overlap and discuss questions under each sphere (L8).

7. Top-down vs bottom-up

Define both approaches from L27. Reproduce the hospital hierarchy for top-down (L28) and the examination-suite aggregation for bottom-up (L29).

5. Coverage Limits and Final Study Priority

Question area	Lecture support	What is still needed
Best structure for a building project	Selection criteria are available (L30-31)	The answer must justify a choice using project size, complexity, integration and authority needs; no single universal answer is stated.
Project team and teamwork	Dedicated teams and matrix overlap (L16-24)	Definitions, team-development theory, functions and teamwork principles require another lecture.
Conceptual design process with top-down design	Top-down design only (L27-28)	The conceptual-design process is not included in this lecture.
Purposes of organizational structure	Hierarchy, authority, coordination and control are implied (L3-4)	A separate formal list of purposes is not explicitly provided.
Weak, balanced and strong matrix	Power relationship is illustrated (L21)	Detailed prose definitions are brief; expand from the chart and comparison table.

Study order	Slides	Focus
Priority 1	L9-26	Functional, projectized and matrix structures; authority, comparisons and pros/cons
Priority 2	L2-6	Meaning, challenges, hierarchy, policy and critical issues
Priority 3	L30-31	Selection criteria and applied structure choice
Priority 4	L32-33	Rapid comparison and summary revision
Priority 5	L7-8 and L27-29	Three Sphere Model plus top-down/bottom-up design

Bottom line: The lecture directly or substantially supports 15 of the 17 linked sub-questions. The highest-value preparation is the comparative study of functional, projectized and matrix structures.